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Superseding
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MILITARY SPECIFICATION

ALUMINUM WOOL

This specification has been approved by the Department of the Air Force and by the Navy Bureau of Naval Weapons.

1. SCOPE

1.1 Scope.- This specification covers various types of aluminum wool.

1.2 Classification.- The aluminum shall be of the following types as specified by the procuring activity.(see 6.2):

Type I - Coarse
Type II - Medium
Type III - Fine

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Boxes, Fiber

Military

MIL-B-10377	Box, Wood Cleated, Veneer, Paper Overlaid

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

FSC 5350

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- The materials shall be of a high quality that will insure the production of aluminum wool suitable to meet the requirements of this specification.

3.2 Construction and physical requirements.-

3.2.1 The aluminum wool shall be made from bar, rod, wire, or tubing of commercial 3S or 56S aluminum alloy.

3.2.2 Strands.- The aluminum wool shall be composed of numerous long-fibered thin strands of aluminum alloy. The strands shall have sharp, smooth cutting edges, and shall be free from chips, short ends, or materials other than aluminum-alloy fiber. The individual strands shall be uniform in texture and size.

3.2.3 Strand or filament sizes.- The strand or filament sizes for each type shall not exceed the values specified in table I.

TABLE I

Permissible strand sizes

Type	Maximum (inch)	Minimum (inch)	Average shall fall within range (inch)
I - Coarse	0.018	0.007	0.016 - 0.009
II - Medium	0.010	0.005	0.009 - 0.006
III - Fine	0.008	0.002	0.007 - 0.003

3.2.4 Contour of strands.- The contour of the strands shall be substantially uniform and free from raggedness, spirals, and whorls. The presence of any of these defects in more than 10 percent of the wool shall be cause for rejection.

3.3 Weight.- The aluminum wool shall not show a weight loss greater than 15 percent when subjected to the brittleness test described in 4.4.4.

3.4 Matting.- The aluminum wool shall not mat or wad easily while in use.

3.5 Workmanship.- All details of workmanship shall be in accordance with high-grade manufacturing practice for this type of product.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility.- Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Classification of tests.- All the tests required herein for the testing of the aluminum wool are classified as acceptance tests, for which necessary sampling techniques and methods of testing are specified in this section.

4.3 Sampling.-

4.3.1 Lot.- For the purpose of inspection, a lot shall consist of 250 pounds or less of each type submitted for acceptance.

4.3.2 Samples.- A sample of not less than 1/4 pound of aluminum wool from each lot represented shall be selected in accordance with Standard MIL-STD-105, using an AQL of 4.0 percent defective. Each sample shall be suitably tagged to identify it with the lot represented. Samples are to be taken and subjected to tests as specified herein.

4.4 Tests.-

4.4.1 Inspection of product.- The aluminum wool shall be subjected to inspection by an authorized Government inspector. When inspection is conducted at the contractor's plant, tests shall be conducted by the contractor under the supervision of the authorized Government inspector.

4.4.2 Measurement.- Not less than 15 measurements of the strand size shall be made on each sample to determine conformance with the size requirements specified in 3.3. The measurements shall be made with a micrometer.

4.4.3 Examination for defects.- The contour of the strands and the examination for defects shall be made at a magnification of 10 to 35 times, using reflected light.

4.4.4 Brittleness test.- Approximately 10 grams of the sample shall be rolled into a ball 2-1/2 inches in diameter, using the palms of the hands. The duration of the rolling action shall be between 40 and 50 seconds. At the end of this time, the ball shall be pulled into an oval shape and the small particles shaken out. The aluminum wool shall not show a weight loss greater than 15 percent when subjected to this test.

4.4.5 Packing and marking.- The inspector shall ascertain that the packing and marking of the aluminum wool conform to this specification.

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4.5 Rejection.- Failure of any sample of the aluminum wool to conform to any part of the requirements of this specification shall be cause for rejection of the entire lot represented. The material shall not be resubmitted for approval without furnishing full particulars concerning previous rejections and measures taken to overcome the defects.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C as specified. (See 6.2.)

5.1.1 Level A.-

5.1.1.1 Unit packaging.- Unless otherwise specified, aluminum wool shall be packaged in folding paperboard boxes or tubes conforming to Specification PPP-B-566, containing either 1/2 pound or 1 pound of wool. No preservation is required.

5.1.2 Level C.-

5.1.2.1 Preservation and packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing.- Packing shall be level A, B, or C as specified. (See 6.2.)

5.2.1 Level A.- The aluminum wool packaged as described in 5.1.1.1 shall be packed in snug-fitting overseas, exterior-type shipping containers conforming to Specification MIL-B-10377 or PPP-B-636. Fiberboard containers shall be not less than 275 pounds Mullen test.

5.2.2 Level B.- The aluminum wool packaged as described in 5.1.1.1 shall be packed in domestic-type shipping containers conforming to Specification PPP-B-601, MIL-B-10377, PPP-B-591, or PPP-B-636.

5.2.3 Level C.- Aluminum wool packaged as specified in 5.1.1.1 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- The aluminum wool covered by this specification is suitable for cleaning and removing corrosion products from aluminum surfaces and for use in air filters.

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6.2 Ordering data.- Procuring documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type desired. (See 1.2.)
(c) Quantity.
(d) Selection of applicable levels of preservation, packaging, and packing. (See section 5.)

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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